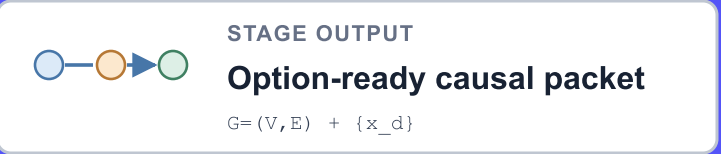
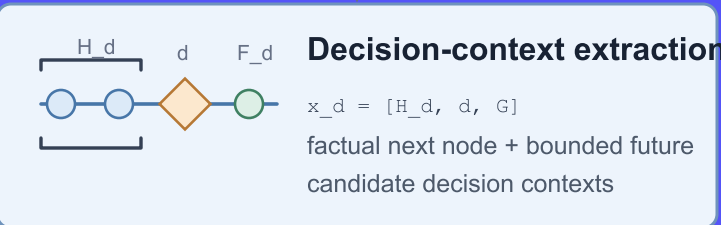
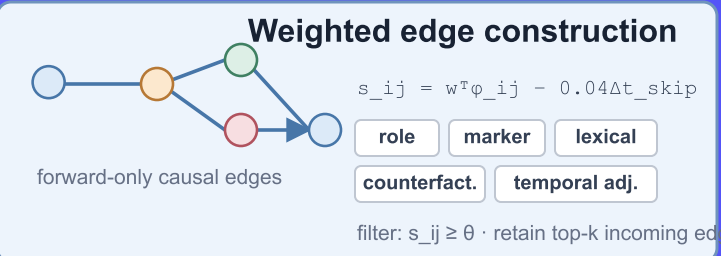
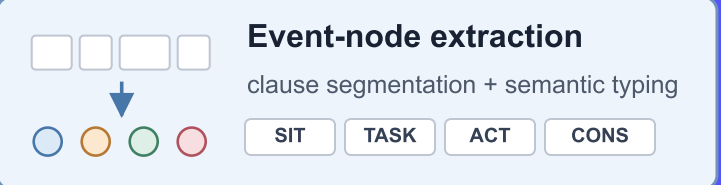
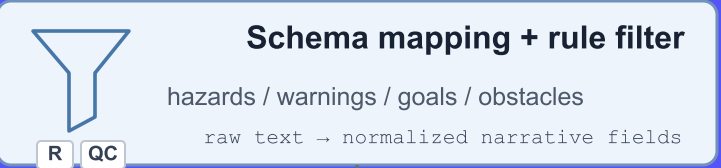


Data Source and DAG Construction

Narratives are normalized into a shared, decision-centered graph representation.

SOURCE CORPORA



Mental Time Travel Options Generation

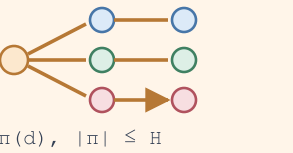
The planner controls path semantics; the LLM realizes only surface-form actions.

DECISION-CENTERED INPUT

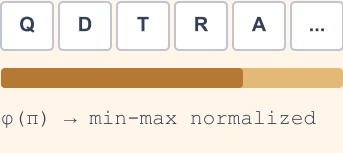


$$x_d = [H_d, d, G]$$

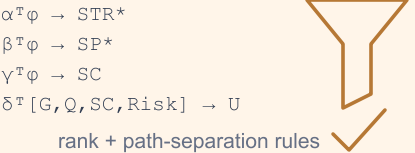
1. Enumerate paths



2. Encode path



3. Weight + filter



FOUR CONTROLLED OPTION MODES

O_fact Factual anchor

Follow the observed next node.

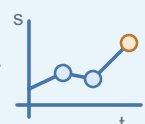
anchor = Next_gold(d)
reference branch; not an optimality claim



O_STR Spatiotemporal reconstruction

Recover time, place, order, constraints.

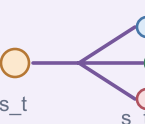
arg max STR*(pi)



O_SP Self-projection

Project future state, resources, risk.

arg max SP*(pi)



O_SC Self-consistency

Match prior goals, roles, constraints.

arg max SC(pi | H_d)



4. Constrained LLM realization

one short action per slot; no score or future leakage

surface text ~ {context, anchor path, generation rule}



VISIBLE MODEL PACKET

Context + four action options



HIDDEN GOLD METADATA

Types + scores + consequence paths

Task Evaluation

Two independent probes use the same benchmark packet.

Task 1 Option-Structure Understanding

INPUT context + all four visible options

	STR	SP	SC	U
o1				
o2				
o3				
o4				

$$\hat{S} \in [0, 1]^{4 \times 4}$$

gold S from paths

MODEL PREDICTION STR / SP / SC / U per option
score every option; do not select an action

ALIGNMENT SA + top-option identification

$$C_d = .30 SA_d + .70 ID_d$$

$$C_STR \cdot C_SP \cdot C_SC \cdot C_U \rightarrow \text{Task 1}$$

Task 2 Fact-Conditioned Consequences

INPUT context + fixed O_fact



STRUCTURED LABELS GA · RE · CV · SEC
+ fine-grained outcome-tag vector

goal + risk ↑ constraint chain effect

HIDDEN GOLD factual DAG continuation

SCORING Macro-F1 + mLOA + Exact Match

$$F_GA \cdot F_RE \cdot F_CV \cdot F_SEC \rightarrow \text{Task 2}$$